eSU in Python: User Guide

Elementary Effects global sensitivity analysis with Enhanced Sampling for Uniformity (eSU)

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1. What the program does

`esu.py` is a small Python program for **global sensitivity analysis** (GSA) with the **Elementary Effects (EE)** method of Morris (1991). It uses the **eSU** sampling strategy (Enhanced Sampling for Uniformity) of Khare et al. (2015) and Chitale et al. (2017). It is a Python port of the group's eSU MATLAB package (Khare and Muñoz-Carpena, 2022).

Global sensitivity analysis tells you which model inputs control a model output when all inputs change together over their full ranges. We call each uncertain input a **factor** (a parameter, a boundary condition, an input value). The EE method is a **screening** method. It needs few model runs. It sorts factors into three groups:

- factors with a negligible effect (you can fix them at a nominal value),
- factors with an important, mostly linear and additive effect,
- factors with an important non-linear effect or with interactions with other factors.

The program does four things:

1. It makes the eSU sample: a set of factor values for your model runs.
2. It maps the sample from the unit scale [0, 1] to the real factor ranges.
3. It calculates the elementary effects and the sensitivity measures μ^*, μ and σ from your model outputs.
4. It draws the two screening plots (μ^*–σ and μ–σ).

You run your model yourself between steps 2 and 3. The model can be a Python function or any external program (Fortran, R, Excel, a GUI model).

2. Files and installation

2.1 Files

The program is on GitHub: https://github.com/carpena1/eSU-py. The program itself is one file, `esu.py`. The other files are examples, tests and this guide.

File	What it is
`esu.py`	The program (a Python module). All the functions are in this file.
`esu_files.py`	A ready-to-use script for models outside Python. It reads and writes CSV files (Section 5).
`examples/example_gfunction.py`	Example A: a complete analysis of a model written in Python (Section 4).
`examples/factors.csv`	Example B: an example factor table for `esu_files.py`.
`examples/demo_external_model.py`	Example B: a stand-in for an external model.
`docs/eSU_Python_Guide.md`, `.pdf`	This guide. `docs/build_pdf.sh` rebuilds the PDF.

File	What it is
`tests/test_esu.py`	Tests of the sampler and of the EE calculation (Section 11).
`tests/validate_esu.py`, `tests/validate_salib.py`	Comparisons with the MATLAB package and with SALib.
`requirements.txt`	The Python packages you need.
`LICENSE`, `CITATION.cff`	MIT license; how to cite.

2.2 Installation

You need Python 3 (3.10 or newer is a safe choice), **numpy** (1.24 or newer) and **matplotlib** (3.6 or newer).

1. Install Python 3 from https://www.python.org (or use Anaconda).
2. Get the program. With git:

   ```
   git clone https://github.com/carpena1/eSU-py.git
   ```

   Without git: open https://github.com/carpena1/eSU-py, click **Code** → **Download ZIP**, and unzip the file.
3. Open a terminal in the `eSU-py/` folder.
4. Make a virtual environment (a private Python installation for this folder):

   ```
   python3 -m venv venv
   ```

5. Activate it. On macOS and Linux:

   ```
   source venv/bin/activate
   ```

   On Windows: `venv\Scripts\activate`.
6. Install the two packages:

   ```
   pip install -r requirements.txt
   ```

7. Run the example to check the installation:

   ```
   cd examples
   python example_gfunction.py
   ```

For your own work, you can also copy `esu.py` (and `esu_files.py`) into your project folder. The example scripts find `esu.py` in the same folder or in the folder one level up.

> **Note.** `import esu` sets the matplotlib "Agg" backend. The plots go to PNG files. `plt.show()` does not open a window after you import `esu`.

3. The method in short

3.1 Factors, levels and the unit scale

You give each factor i a range $[lo_i, hi_i]$ with a **uniform** distribution. The program works on the **unit scale**: it maps each range to [0, 1]. On this scale each factor can take p equally spaced **levels**:

$$x \in \left\{0, \frac{1}{p-1}, \frac{2}{p-1}, ..., 1\right\}$$

p must be even (the usual values are 4, 6, 8, …, 16).

3.2 Trajectories

The EE method uses **trajectories** (Morris, 1991). A trajectory starts at a random point on the grid. Then it changes **one factor at a time**, in a random order, by a fixed step Δ:

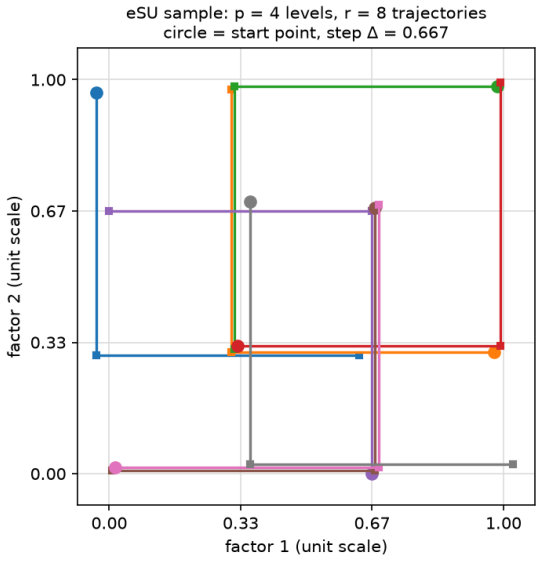
$$\Delta = \frac{p}{2(p-1)}$$

If the factor value is below 0.5, the step is $+\Delta$. Otherwise it is $-\Delta$. This keeps all points on the grid. Each factor moves exactly once, so one trajectory has $d + 1$ points for d factors. A sample of r trajectories needs

$$N = r\,(d + 1) \text{ model runs.}$$

p	**4**	**6**	**8**	**10**	**12**	**16**
Δ (unit scale)	0.667	0.600	0.571	0.556	0.545	0.533

On the real scale the step is $\Delta\,(hi_i - lo_i)$.



An eSU sample with 2 factors, p = 4 levels and r = 8 trajectories. Each colour is one trajectory. Small shifts separate trajectories that overlap. Each factor moves once per trajectory. Each of the 4 levels is a start value exactly twice for each factor.

3.3 What eSU adds: uniform start points

Classical Morris sampling picks the start points at random. With few trajectories, some levels can then be sampled many times and others not at all. **eSU** fixes this. For each factor, each of the p levels is a start value exactly r/p times, in random order (Khare et al., 2015; Chitale et al., 2017). So the sample covers each factor range uniformly, also with a small r. This is why r **must be a multiple of** p (and even).

3.4 Elementary effects and the sensitivity measures

In each trajectory, factor i moves once, from point $\boldsymbol{x}$ to point $\boldsymbol{x}'$. Its **elementary effect** is the change in the output divided by the step (both on the unit scale):

$$EE_i = \frac{Y(\boldsymbol{x}') - Y(\boldsymbol{x})}{x_i' - x_i} = \pm\frac{Y(\boldsymbol{x}') - Y(\boldsymbol{x})}{\Delta}$$

Because the step is on the unit scale, EE_i has the **units of the output**. It is about the change in Y over the full range of factor i. So you can compare the EEs of factors with different units.

With r trajectories, each factor has r elementary effects. The program calculates three measures for each factor:

Measure	Formula	Meaning
μ^*	$\frac{1}{r}\sum_k \lvert EE_i^{(k)}\rvert$	Overall importance (Campolongo et al., 2007). Large μ^*: important factor.
μ	$\frac{1}{r}\sum_k EE_i^{(k)}$	Mean effect, with its sign. Positive: Y increases when the factor increases. Effects of opposite sign cancel.

Measure	Formula	Meaning
σ	$\sqrt{\frac{1}{r-1}\sum_k \left(EE_i^{(k)} - \mu\right)^2}$	Spread of the effects. Large σ: the effect changes with the factor value (non-linear) or with the values of other factors (interaction).

The EE method cannot separate non-linearity from interactions. A large σ means “non-linear **or** interacting **or** both”.

4. Example A: a model written in Python

Use this path when you can call your model as a Python function. The file `examples/example_gfunction.py` has the complete script. It uses the **Sobol g-function**, a standard test model with a known answer:

$$Y = \prod_{i=1}^{6} \frac{|4x_i - 2| + a_i}{1 + a_i}, \quad x_i \in [0,1], \quad a = (0,\ 1,\ 4.5,\ 9,\ 99,\ 99)$$

A small a_i makes factor i important. So the true order is x1 > x2 > x3 > x4, and x5, x6 are negligible.

The script has six steps:

```python
import numpy as np
import esu

# 1. Factors and their ranges (uniform distributions)
names = ["x1", "x2", "x3", "x4", "x5", "x6"]
lo = [0.0] * 6
hi = [1.0] * 6
d = len(names)
a = np.array([0.0, 1.0, 4.5, 9.0, 99.0, 99.0])

# 2. The model: one output value for one row of factor values
def model(x):
    return np.prod((np.abs(4 * x - 2) + a) / (1 + a))

# 3. Sampling design
p, r = 8, 32                                  # levels, trajectories
rng = np.random.default_rng(12345)            # seed -> repeatable sample
unit = esu.esu_sample(d, r, p, rng)           # r*(d+1) rows x d columns, in [0, 1]
real = esu.map_uniform(unit, lo, hi)          # real factor values

# 4. Run the model for every row
Y = np.array([model(row) for row in real])

# 5. Elementary effects and the sensitivity measures
EE = esu.elementary_effects(unit, Y, p)       # d x r
mu_star, mu, sigma = esu.ee_measures(EE)
print(esu.format_measures(names, mu_star, mu, sigma))

# 6. Screening plot
```

```
esu.ee_plot(mu_star, mu, sigma, EE, names, "g-function", "gfunction_EE.png",
            rng=np.random.default_rng(0))
```

To use it for your own model:

1. Change `names`, `lo` and `hi` (step 1).
2. Replace `model(x)` with a call to your model (step 2). `x` is one row of real factor values, in the order of `names`. The function must return one number.
3. Set `p` and `r` (step 3; see Section 6).

The program writes this table (`gfunction_EE_Measures.txt`):

```
facname          mu*           mu        sigma  Monotonic  Interactions
x1        2.1682e+00   4.9419e-02   2.5781e+00      N           Y
x2        1.0449e+00   1.9141e-01   1.3586e+00      N           Y
x3        4.1894e-01   2.5309e-02   5.8991e-01      N           Y
x4        2.5800e-01  -4.6759e-03   3.5399e-01      N           Y
x5        2.3341e-02   4.9665e-03   3.3560e-02      N           Y
x6        1.8526e-02   4.0030e-03   2.3262e-02      N           Y
```

Section 7 explains how to read the table and the plot.

5. Example B: a model outside Python (CSV files)

Use `esu_files.py` when your model is a separate program. You do not need to write Python. The data goes through CSV files (comma-separated text files that Excel can open and save).

> **Where to run the commands.** Run all commands in your work folder: the folder that has `factors.csv`. Python must find `esu_files.py`, so write its path before the name:
>
> - In the `examples/` folder, `esu_files.py` is one level up. Use `python ../esu_files.py ...` (as in the commands below).
> - If you copy `esu_files.py` and `esu.py` into your work folder, use `python esu_files.py ...`.
> - From any other folder, use the full path, for example `python ~/eSU-py/esu_files.py ...`.

5.1 Step 1: describe the factors

Make a file `factors.csv` with a header row and one row per factor. The columns are `name`, `lo` and `hi`:

```
name,lo,hi
Ks,1,50
porosity,0.30,0.50
CN,60,90
slope,0.01,0.10
dummy,0,1
```

> **Tip.** Add a **dummy** factor that your model does not use. Its μ^* must be zero (or close to zero). Factors with a μ^* near the dummy value are negligible.

5.2 Step 2: make the sample

```
python ../esu_files.py sample factors.csv --p 8 --r 16 --seed 1
```

The program writes two files:

- `esu_sample.csv`: the factor values in real units. It has one row per model run ($N = r(d + 1)$ rows) and one column per factor.
- `esu_design.json`: p, r, the factor names and the ranges. Step 4 reads this file. Do not change it.

Option	Default	Meaning
`--p`	8	Number of levels. Must be even.
`--r`	16	Number of trajectories. Must be a multiple of `p`.
`--seed`	none	Random seed. The same seed gives the same sample. Without a seed, each call gives a new sample.
`--out`	`esu_sample.csv`	Name of the sample file.
`--design`	`esu_design.json`	Name of the design file.

5.3 Step 3: run your model

1. Run your model once for each row of `esu_sample.csv`.
2. Keep the **same row order**. Do not sort or delete rows.
3. Save the results in `esu_outputs.csv`: a header row with the output names, then one row per model run and one column per output. For example:

```
runoff,peak
12.81,3.204
13.07,3.268
...
```

If a run fails, leave the cell empty or write `nan` or `NA`. The program then ignores the elementary effects that use this run (see Section 8.4).

> **Try it.** The `examples/` folder has a complete Example B. The script `demo_external_model.py` acts as the external model. From `examples/`:
>
> ```
> python ../esu_files.py sample factors.csv --p 8 --r 16 --seed 1
> python demo_external_model.py
> ```

```
python ../esu_files.py analyze
```

The model does not use `porosity` and `dummy`, so their μ^* is zero.

5.4 Step 4: analyze

```
python ../esu_files.py analyze
```

The program prints the measures (sorted by μ^*) and writes, for each output column:

File	Contents
`EE_Measures.txt`	The measures table for all outputs (same layout as the MATLAB `*_EE_Measures.txt`).
`EE_raw_<output>.csv`	All elementary effects: one row per factor, one column per trajectory.
`EE_<output>.png`	The two screening plots.

Options: `--outputs` (default `esu_outputs.csv`), `--sample` (default: the file named in the design file), `--design` and `--report` (default `EE_Measures.txt`).

6. Choosing p and r

The cost of the analysis is $N = r\,(d + 1)$ model runs. So r sets the cost.

Choice	Rule	Common start value
p (levels)	Even number, 4 to 16	8
r (trajectories)	Even and a multiple of p	16 to 32 (more for many factors or a noisy model)

Example: 20 factors, $p = 8$, $r = 16$ gives $16 \times 21 = 336$ runs.

Check the result. The eSU sample is random. Do this check before you trust a factor ranking:

1. Repeat the analysis with two or three different seeds.
2. Compare the order of the important factors and the confidence intervals in the plots.
3. If the order of the important factors changes, increase r (for example from 16 to 32) and repeat.

The negligible factors often change order among themselves. That is normal and not important.

7. Reading the results

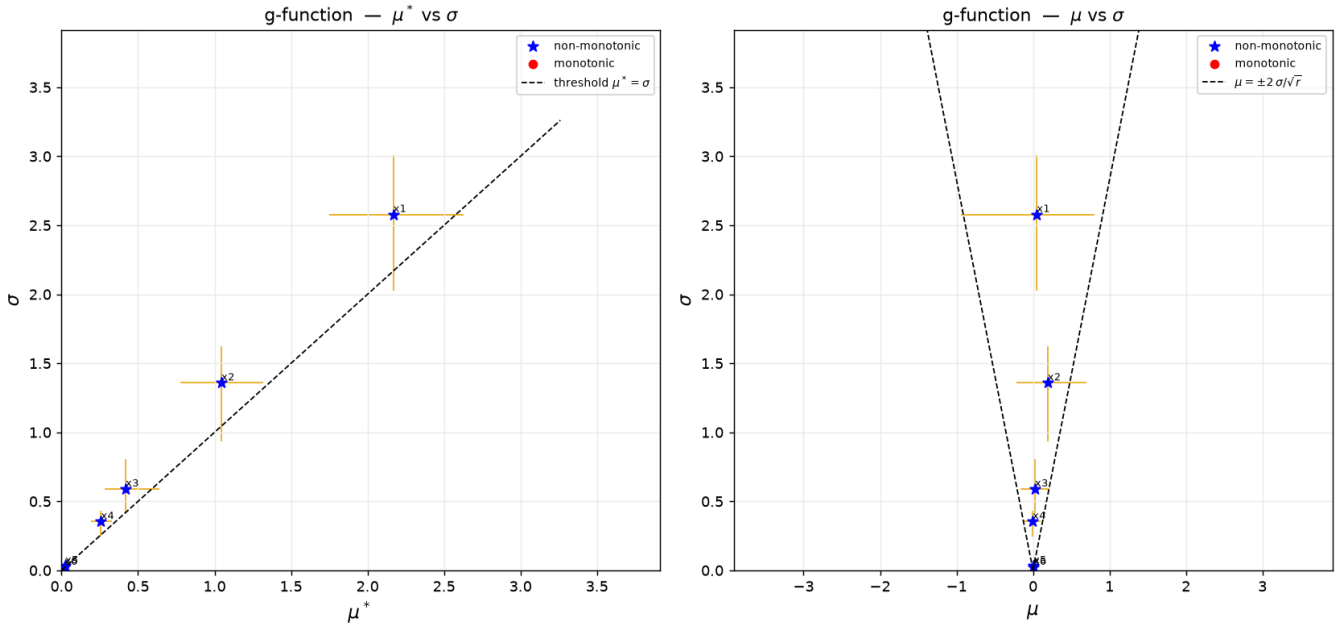
7.1 The measures table

Column	Meaning
`mu*`	μ^*: overall importance. Use it to rank the factors.
`mu`	μ: mean effect with sign.
`sigma`	σ: non-linearity and/or interactions.
`Monotonic`	`Y` when $\mu^* = \lvert\mu\rvert$ exactly. This means all elementary effects of the factor have the same sign: in all trajectories, Y changed in the same direction.
`Interactions`	`Y` when $\sigma \geq \mu^*$. The spread of the effects is as large as their mean size. The factor is non-linear and/or interacts with other factors.

Two cautions:

- `Monotonic = Y` is true only for the sampled steps of size Δ. It is not a proof that the model is monotonic.
- A factor with no effect has $\mu^* = \mu = \sigma = 0$. The program then writes `Y` in both flag columns. Ignore the flags for such a factor.

7.2 The screening plots



Screening plots for the g-function example (p = 8, r = 32, 224 runs). Left: μ* against σ. Right: μ against σ. Red circles: monotonic factors. Blue stars: non-monotonic factors. Golden bars: bootstrap 95% confidence intervals.

Left plot, μ^* against σ.

- The distance from the origin shows the overall importance. Factors near the origin are negligible.
- The dashed line is $\mu^* = \sigma$.
- Factors **below** the line have a mostly linear and additive effect.
- Factors **above** the line have a mostly non-linear effect and/or interactions.

Right plot, μ against σ.

- The two dashed lines are $\mu = \pm 2\sigma/\sqrt{r}$, that is, two standard errors of the mean (Morris, 1991).
- Factors **outside** the wedge (to the left or right of the lines) have a mean effect that is clearly not zero. Their effect has one main direction: positive ($\mu > 0$) or negative ($\mu < 0$).
- Factors **inside** the wedge have a mean effect that is not different from zero, because effects of opposite sign cancel. A factor inside the wedge with a large σ is important but non-monotonic and/or interacting.

Confidence intervals. The golden bars are bootstrap 95% confidence intervals for μ^*, μ and σ (100 resamples of the r elementary effects). Wide bars mean the estimate is uncertain. If the bars of two factors overlap a lot, the sample cannot tell which one is more important. Increase r if you need that order.

Labels. By default the plot labels all factors. With many factors, use `n_label` (label the top factors by μ^*) or `label_mask` (label a chosen set). See Section 9.

7.3 The g-function example

The results agree with the known answer:

- The order by μ^* is x1 > x2 > x3 > x4. The values of x5 and x6 are about 100 times smaller than that of x1. You could fix x5 and x6.
- All factors are non-monotonic ($\mu \approx 0$, blue stars inside the wedge). This is correct: the term $|4x - 2|$ has a V shape, so Y goes down and then up.
- All factors are above the $\mu^* = \sigma$ line. This is correct: the product form makes the effect of each factor depend on the other factors (interactions).

8. Practical tips

8.1 Factors that are not uniform

`esu.py` supports **uniform** ranges only.

For a factor that spans orders of magnitude (for example a hydraulic conductivity from 0.1 to 100), sample its logarithm:

1. Use the factor `log10_Ks` with the range $[\log_{10}0.1, \log_{10}100] = [-1, 2]$.
2. In your model (or before you run it), use `Ks = 10**log10_Ks`.

For other distributions (normal, triangular, discrete, categorical), use the MATLAB package (Section 10).

8.2 More than one output

Use the same sample for all outputs. Call `elementary_effects` and `ee_measures` once for each output (`esu_files.py` does this for each column).

The measures have the units of the output. So compare factors **within** one output. To compare outputs, divide the measures by a reference value of each output, for example its value in a base run or its standard deviation.

For a time series output (for example daily flow), you can repeat the analysis for each time step, or use a summary of the series (a total, a peak, a goodness-of-fit value).

8.3 Ranking and fixing factors

Rank the factors by μ^*, or by the distance from the origin of the μ^*–σ plot, $\sqrt{\mu^{*2} + \sigma^2}$. The second choice gives credit to both the main effect and the interactions.

To decide which factors to fix, compare them with a reference. Two common choices are:

- the μ^* of the dummy factor (Section 5.1), or
- a percentage of a reference output value. For example, the calibration tool of the CUENCA hydrologic model keeps a factor when $100\sqrt{\mu^{*2} + \sigma^2}\ /|Y_{base}| \geq 1\%$.

Keep the threshold low. It is better to keep a factor of small importance than to fix a factor that matters.

8.4 Failed model runs

Some runs can fail, because the sample also tests extreme combinations of factor values. Give the failed runs the value `NaN` (`np.nan` in Python; an empty cell or `nan` in `esu_outputs.csv`).

- A failed point removes the elementary effects that use it: one effect for the first or last point of a trajectory, two effects for a point in the middle.
- The measures use the remaining effects.
- A few failures are acceptable. Many failures make the estimates uncertain. Then check the model, or reduce the factor ranges.

8.5 Reproducible results

- Give a seed to the sampler: `esu.esu_sample(d, r, p, np.random.default_rng(1))`, or `--seed 1` in `esu_files.py`.
- Give a seed to the plot too (`rng=` in `ee_plot`). The bootstrap intervals are random.
- Save the sample file with your results. The measures depend on the exact sample.

8.6 Faster runs

The model runs are independent. You can run them in parallel on all the cores of your computer. For a Python model:

```
from concurrent.futures import ProcessPoolExecutor

if __name__ == "__main__":
    with ProcessPoolExecutor() as ex:          # one process per core
        Y = np.array(list(ex.map(model, real, chunksize=8)))
```

`ex.map` keeps the row order, so `Y` matches the sample. Define `model` at the top level of the file. Put the sampling and the analysis under `if __name__ == "__main__":` too, because on Windows and macOS each new process reads the file again. For external models, run several copies of the model at the same time, each in its own folder.

9. Function reference

All functions are in `esu.py`. “Unit” means the [0, 1] scale. `rng` is a numpy random generator (`np.random.default_rng(seed)`); without it, the result changes from call to call.

9.1 Sampling

Function	Returns
`esu_sample(d, r, p, rng=None)`	The eSU sample on the unit scale: an array with $r(d+1)$ rows and d columns. Rows $k(d+1)$ to $k(d+1)+d$ are trajectory k. Raises `ValueError` if r is odd or not a multiple of p.
`map_uniform(unit, lo, hi)`	The sample in real units: `lo + (hi - lo) * unit`, for each column.
`esu_delta(p)`	The step $\Delta = p/(2(p-1))$.
`esu_levels(p)`	The p grid levels $0, 1/(p-1), \ldots, 1$.

Lower-level functions (used by `esu_sample`): `esu_pgen(r, p, rng)` (uniform start levels for one factor), `esu_base_points(d, r, p, rng)` (start points of all trajectories), `esu_perturb_order(d, rng)` (random order of the factor moves), `esu_traj(base_row, order, p)` (one trajectory).

9.2 Measures

Function	Returns
`elementary_effects(sample, Y, p)`	The EE matrix: d rows (factors) and r columns (trajectories). `sample` can be on the unit or the real scale (the program uses only the direction of each step). `Y` has one value per sample row, in the same order. Non-finite `Y` values give `NaN` effects.
`ee_measures(EE)`	The arrays `mu_star`, `mu`, `sigma`, one value per factor. `NaN` effects are ignored.
`monotonic_mask(mu_star, mu)`	`True` where $\mu^* = \lvert\mu\rvert$.
`format_measures(labels, mu_star, mu, sigma)`	The measures table as text (Section 7.1).
`bootstrap_ci(x, stat, n=100, alpha=0.05, rng=None)`	A percentile bootstrap confidence interval (low, high) of `stat` over the finite values of `x`.

9.3 Plot

`ee_plot(mu_star, mu, sigma, EE, labels, title, out_png, n_label=None, label_mask=None, rng=None)`

Draws the two screening plots of Section 7.2 and saves them to `out_png`.

- `n_label`: label only the `n_label` factors with the largest μ^*.
- `label_mask`: an array of `True`/`False`, one per factor. Label only the factors with `True`. It overrides `n_label`.
- `rng`: random generator for the bootstrap intervals.

10. Differences from the MATLAB package

The Python program gives the same measures as the MATLAB package (Khare and Muñoz-Carpena, 2022) for the same sample and outputs. The test `tests/validate_esu.py` compares them on a MATLAB example; the relative difference of μ^* and σ is about 3×10^{-5}, and the factor order is the same. (To run it, get the MATLAB package and set the environment variable `ESU_MATLAB_DIR` to its `EE_Sensitivity_Measures` folder.) The differences are:

Topic	MATLAB package	`esu.py`
Distributions	Uniform, log-uniform, normal, triangular, discrete, categorical	Uniform only (Section 8.1)
Samplers	eSU and other trajectory samplers, radial eSU	eSU trajectories only
Input	`.fac` factor file	Python lists, or `factors.csv` with `esu_files.py`

Topic	MATLAB package	`esu.py`
Time series measures and plots	Yes	No (Section 8.2)
Samples	Random	Random, with the same properties (not the same numbers)
Plot confidence intervals	Bootstrap, BCa	Bootstrap, percentile (the measures are exact; only the error bars differ a little)

11. Tests

To check the program, run the tests from the `eSU-py/` folder:

```
python tests/test_esu.py
```

The five tests check that:

1. each level is a start value the same number of times (eSU uniformity),
2. the sample has the right size and stays in [0, 1],
3. each step moves exactly one factor by Δ, and each factor moves once per trajectory,
4. `map_uniform` maps 0 to `lo` and 1 to `hi`,
5. for a linear model, μ^* is the true effect and $\sigma = 0$.

All five must show `PASS`.

12. How to cite

When you use this program in a report or a paper, cite the method papers:

- Khare et al. (2015) and Chitale et al. (2017) for the eSU sampling,
- Morris (1991) and Campolongo et al. (2007) for the EE method and μ^*,
- Khare and Muñoz-Carpena (2022) for the original MATLAB software.

Khare et al. (2019) shows an application to large hydrologic and water quality models.

References

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